# Industrial Process Modeling and Monitoring Based on Jointly Specific and Shared Dictionary Learning

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***Abstract*— With the development of industrial cyber–physical systems (ICPS), data of industrial processes are collected and used for data-driven process monitoring. However, due to the variation of working conditions, the data samples are always featured by characteristics and commonalities. Generally, in industrial process data, characteristics are sparse, and commonalities are strongly correlated. The presence of characteristics hampers the feature extract of industrial data, thus further bringing difficulties to process monitoring. To solve this problem, this article proposes a jointly specific and shared dictionary learning (JSSDL) method. Specifically, we first build a specific dictionary and a shared dictionary to reconstruct characteristics and commonalities feature, respectively. Then, considering the sparsity of characteristics and the strong correlation of commonalities, we add sparsity constraints to specific dictionaries and low-rank constraints to a shared dictionary. After that, an iterative optimization algorithm is proposed to optimize the two dictionaries. When online data samples arrive, we use the two dictionaries to reconstruct the data and determine the data that are normal or faulty according to the reconstruction error. To verify the superiority of the proposed JSSDL method in process monitoring, we do extensive experiments. The experimental results demonstrate that the proposed method can achieve satisfactory monitoring results compared with several state-of-the-art methods.**

***Index Terms*— Dictionary learning, low-rank and sparse, process monitoring, shared dictionary, specific dictionary.**

## I. Introduction

WITH the rapid development of information technologies, the information interaction between production processes and cyberspace is increasingly intensive, which promotes the application of industrial cyber–physical systems (ICPS) [1], [2] in industrial processes. Generally, a complete ICPS consists of two layers: the physical layer and the cyberspace layer. In the physical layer, the factory completes production equipment operating and product manufacturing. In cyberspace, the data is collected and stored, and monitoring of the state of industrial processes is accomplished by some advanced data analysis methods. The widespread use of ICPS in industrial processes has made process monitoring increasingly convenient and efficient.

As the production process becomes increasingly complex, a small fault may cause incalculable damage to the whole system [3], [4]. Process monitoring can obtain the real-time status of the production process, and when abnormalities are monitored to occur, an alarm can be triggered to eliminate safety hazards and reduce economic losses. As a result, process monitoring is significant to maintain the stable and safe operation of industrial production. Generally, process monitoring methods include model-based methods and data-driven methods. The former builds the model of industrial process based on the first principle of the industrial process; the latter monitors the industrial process based on extracting the underlying features from the measurement data. Data-driven methods without the need of exact first principle knowledge so that they are suitable for modern complex industrial systems and, thus, have been widely studied and applied [5], [6]. Typical data-driven methods include principal component analysis (PCA), partial least squares (PLSs), [7], [8], and so on. Recently, with the development of artificial intelligence, some machine learning and deep learning methods have been used for industrial process modeling and monitoring, such as the slow feature analysis method [9], the Bayesian analysis method [10], and the neural network method [11]. Recently, there have been some important advances in the field of process monitoring. Si *et al.* [12] first proposed a pioneering method to accurately decompose measurements into KPI-related and KPI-unrelated parts, which achieves excellent performance in nonlinear process monitoring. Chen *et al.* [13] proposed a just-in-time-learning-aided canonical correlation analysis method for multimode process monitoring and solved the limitations of CCA in handling processes with multiple operating points. Dong and Qin [14] considered the dynamic features of the data and proposed a novel dynamic PCA algorithm to extract explicitly a set of dynamic latent variables with which to capture the most dynamic variations in the data. Jiang *et al.* [15] proposed a distributed computing framework of local–global modeling for nonlinear plant-wide process monitoring. These works have contributed to the development of process monitoring by proposing innovative and effective solutions that address different problems in the field of process monitoring.

As an efficient machine learning method, dictionary learning, which represents the data as a linear combination of atoms, has been fully developed and applied. Due to its good generalization and data representation ability, dictionary

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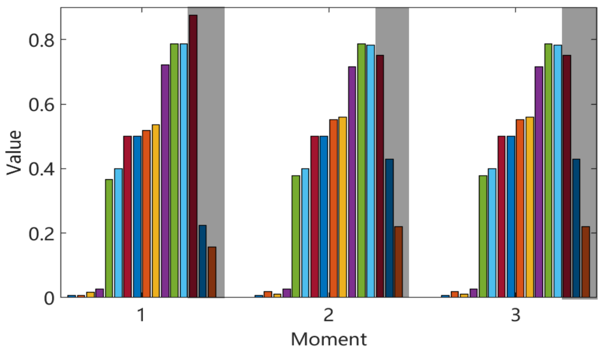



Fig. 1. Visual interpretation of the existence of characteristics and commonalities in industrial data. There are three moments of data in total.

learning has been widely used in image processing, pattern recognition [16], and so on. Dictionary learning has been continuously developed to achieve better feature representation performance, such as the label consistent K-SVD method [17], the supervised dictionary learning method [18], and the discriminative K-SVD method [19]. Likewise, dictionary learning is also applicable to process monitoring. Ren and Lv [20] proposed a fault detection method for semiconductor manufacturing processes based on dictionary learning. Huang *et al.* [21] proposed a multimode process monitoring method based on structured dictionary learning and got good results in the aluminum electrolysis industry. Huang *et al.* [22] presented a discriminative dictionary learning for high-dimensional process monitoring.

Although the abovementioned methods work well for data representation, there are still some difficulties when it is applied to some complex industrial process monitoring. Based on an in-depth analysis of industrial data, we believe that industrial data consist of commonalities and characteristics. Ideally, industrial data would be strongly correlated because of the presence of commonalities. However, due to the variation of working conditions, industrial process data always have the feature of characteristics, which are pervasive and different from each other. In addition, since the characteristics features are not particularly remarkable, they may affect only a few dimensions of raw data. To illustrate the situation vividly, we analyze a dataset of real industrial processes. After normalizing the data, we use three consecutive moments and draw the bar chart, as shown in Fig. 1. There are three clusters in Fig. 1, for each representing one moment of data. The data at one moment have 16 dimensions and a dimension representing industrial data collected on a sensor. We measure the similarity between the data of different moments based on the standard deviation of each dimension. The standard deviations of the first 13 dimensions are less than 0.02. For the last three dimensions, they are 0.0722, 0.1178, and 0.0366, which are much larger than the first 13 dimensions. A larger standard deviation indicates a larger variation in a dimension of different moments. By analyzing the standard deviations, it can be found that the first 13 dimensions of different moments are virtually unchanged. Therefore, we assume that there are strongly similar components in the data at different moments. The standard deviation of the last three dimensions is relatively large. We assume that there are a few components with large variations in the data at different moments. Furthermore, we define strongly similar components in the data as commonalities and components with large variations as characteristics.

Based on the above in-depth analysis of industrial data, we are inspired and developed assumptions about industrial data. We assume that complex industrial process data can be decomposed into commonalities and characteristics. Commonalities feature is strongly correlated, and characteristics feature is sparse. In these assumptions, we decompose industrial process data into two components: commonalities and characteristics, which correspond to some common information and unique information in a set of industrial data, respectively. Commonalities are similar because they represent common information in industrial data, and therefore, commonalities feature behaves as strong correlations. Characteristics represent unique information in industrial data, which are generally sparse because unique information appears in a few dimensions of the data.

When traditional machine learning methods are used to monitor this kind of data, such as in Fig. 1, it may incorrectly regard normal data at moment 1 as a fault. The misjudgment occurs because of the presence of characteristics in moment 1. It is obvious that, in this case, some traditional methods are no longer applicable. Certainly, some studies have taken these problems into account and proposed solutions. Since characteristics exist only in a few dimensions, these methods consider the characteristics as sparse outliers, use some robust approaches to remove them, and then train the model with a data-driven method, such as robust PCA and robust dictionary learning [23]. However, there are also some problems with these methods. First, these methods remove the characteristics in training data as outliers and consider only the commonalities, which causes it cannot learn the characteristics feature of the data, so the performance of process modeling and monitoring is limited. Second, these methods many even face the problem of whether to remove the characteristics from online monitoring data. If not removed, these methods tend to treat the characteristics in normal data as faults, which causes a high false alarm rate (FAR). However, if removed, it may remove the faults of the faulty data as characteristics and, thus, further reduce the fault detection rate (FDR).

Motivated by the situation that industrial data often have both characteristics and commonalities feature, this article proposes a jointly specific and shared dictionary learning (JSSDL) method, which builds a specific dictionary and a shared dictionary, respectively, for characteristics and commonalities feature simultaneously, and solves the process modeling and monitoring problem of the industrial site under the framework of ICPS. The main contribution points of this article are given as follows. First, a task-oriented process monitoring method is proposed to consider the existence of both characteristics and commonalities in industrial data, which is applicable to a wide range of industrial processes. Second, the proposed JSSDL method has a good modeling capability. Since the sparsity of specific dictionaries and sizes of two dictionaries can be adjusted, it has good data representation capability for various complex data. Finally, through the analysis of industrial process data, we verified the existence of characteristics and commonalities. The superiority of the proposed method for process monitoring is verified by compared with some traditional methods.

The rest of this article is structured as follows. In Section II, the process data modeling, optimization algorithm, and steps of process monitoring are given. In Section III, three experiments are given to illustrate the effectiveness of the proposed method. Finally, the conclusion is given in Section IV.

## II. Methodology

### A. Basics of Dictionary Learning

The philosophy of dictionary learning is to linearly represent data by a small number of dictionary atoms, and the atoms are the columns of the dictionary matrix. Mathematically,

$$Y \approx DX \tag{1}$$

where $Y \in R^{n\times m}$, $D \in R^{n\times k}$, and $X \in R^{k\times m}$ denote the data sample matrix, the dictionary, and the sparse coding, respectively. $n$ denotes the dimensionality of the data sample, $m$ denotes the number of samples, and $k$ denotes the number of atoms. To train the dictionary from the data samples, the optimization problem of dictionary learning is defined as follows:

$$\min_{D,X} \|Y - DX\|_F^2, \quad \text{s.t. } \forall i, \ \|x_i\|_0 \le T_0 \tag{2}$$

where $\|Y - DX\|_F^2$ is the reconstruction error term, which ensures the representation capabilities of dictionaries to data. $\|\cdot\|_F$ denotes the Frobenius norm of the matrix. If $A = [a_{ij}]_{m\times n}$, $\|A\|_F = \sqrt{\sum_{i,j} a_{ij}}$. $x_i$ denotes the $i$th column of sparse coding $X$. The constraint of $L_0$ norm ensures the sparsity of the matrix. $\|x_i\|_0$ indicates that few atoms should be used for data representation. Specifically, by approximately replacing $L_0$ norm by the $L_1$ norm, the constrained optimization problem can be transformed into an unconstrained optimization problem as follows:

$$\min_{D,X} \left\{ \|Y - DX\|_F^2 + \lambda \|X\|_1 \right\}. \tag{3}$$

Since there are two optimization variables $D$ and $X$ in (3), and they are not joint convex, the alternate iteration method is introduced to solve this kind of problem, which solves the optimization problem by optimizing all variables one by one, fixing the other variables while updating the current one. In this problem, there are two variables $D$ and $X$. For the sparse coding $X$, the orthogonal matching pursuit (OMP) [24] is often be used. The commonly used methods for solving dictionary $D$ are the method of optimal directions (MODs) and the K-SVD method [25].

### B. JSSDL-Based Process Monitoring

In this section, we introduce the proposed JSSDL method in detail. Generally, the proposed method includes three stages: off-line modeling, optimization algorithm, and online process monitoring. First, we establish a mathematical model for industrial data elaborately. Then, a novel optimization algorithm is designed to train the model. Finally, we use learned dictionaries to reconstruct the online data and perform process monitoring tasks. The framework of the proposed JSSDL method for process monitoring is shown in Fig. 2.

*1) Off-Line Modeling:* According to the assumptions, the industrial data can be modeled as follows:

$$Y = Y_1 + Y_2 \tag{4}$$

where $Y$ denotes the raw data, $Y_1$ denotes the characteristics, and $Y_2$ denotes the commonalities. Based on the characteristics and commonalities feature, $Y_1$ is a sparse matrix, and the columns of $Y_2$ are strongly correlated with each other.

When only the characteristics $Y_1$ are considered, use the specific dictionary $D_1$ to represent $Y_1$. Mathematically, $Y_1 \approx D_1 X_1$. $X_1$ is the sparse coding matrix. According to the characteristics feature, most elements in $Y_1$ are 0. In traditional dictionary learning, each element of data reconstructed by dictionary atoms is a nonzero value. Although the reconstruction error is guaranteed to be small, the sparsity of the data cannot be reconstructed, which is obviously unreasonable. Therefore, sparsity constraint should be added to dictionary learning to ensure that dictionary atoms can extract the sparse feature of the data. The improved optimization problem is formulated as follows:

$$\min_{D_1,X_1} \left\{ \|Y_1 - D_1 X_1\|_F^2 + \lambda_1 \|X_1\|_1 + \lambda_2 \|D_1\|_1 \right\} \tag{5}$$

where $D_1$ denotes the specific dictionary. An $L_1$ norm constraint to $D_1$ can ensure the sparsity of the learned dictionary. When a small number of atoms are used to reconstruct the characteristics, the sparsity of the data can be maintained while minimizing reconstruction error.

When only the commonalities $Y_2$ are considered, use the shared dictionary $D_2$ to represent $Y_2$. Mathematically, $Y_2 \approx D_2 X_2$. $X_2$ is the sparse coding matrix. The strong correlation of $Y_2$ makes $D_2$ to be low-rank. Therefore, a low-rank constraint should be added on $D_2$. Accordingly, the optimization problem is given as follows:

$$\min_{D_2,X_2} \left\{ \|Y_2 - D_2 X_2\|_F^2 + \lambda_3 \|X_2\|_1 + \lambda_4 \operatorname{rank}(D_2) \right\}. \tag{6}$$

Since the kernel norm is a convex approximation of low-rank constraint, (6) can be approximated as follows:

$$\min_{D_2,X_2} \left\{ \|Y_2 - D_2 X_2\|_F^2 + \lambda_3 \|X_2\|_1 + \lambda_4 \|D_2\|_* \right\}. \tag{7}$$

The raw data can be reconstructed by the specific dictionary and shared dictionary. Mathematically,

$$Y \approx D_1 X_1 + D_2 X_2 \tag{8}$$

where $X_1$ and $X_2$ are the coding matrices of the specific dictionary and the shared dictionary. In summary, considering both characteristics and commonalities, the optimization problem of JSSDL can be formulated as follows:

$$\min_{D_1,X_1,D_2,X_2} \left\{ \begin{array}{l} \|Y - D_1 X_1 - D_2 X_2\|_F^2 + \lambda_1 \|X_1\|_1 \\ \quad + \lambda_2 \|D_1\|_1 + \lambda_3 \|X_2\|_1 + \lambda_4 \|D_2\|_* \end{array} \right\}. \tag{9}$$

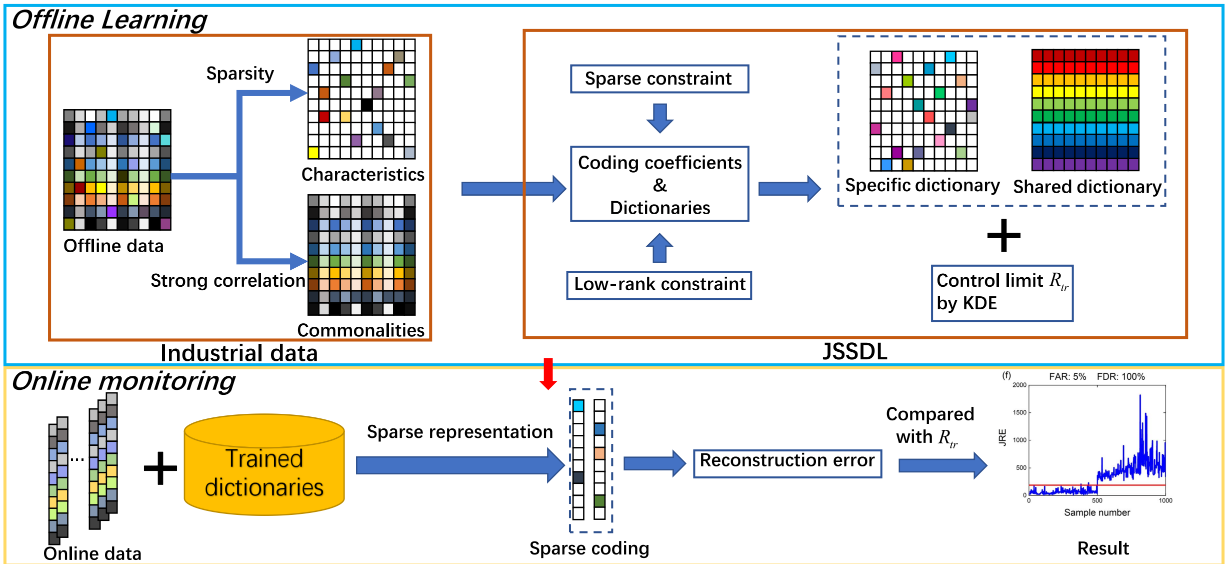



Fig. 2. JSSDL-based process monitoring framework.

Unlike the traditional dictionary learning that uses a single dictionary to represent data, JSSDL uses a specific dictionary and a shared dictionary to represent data and adds a sparsity constraint to the specific dictionary and a low-rank constraint to the shared dictionary, which can deal with industrial data with the complex feature.

After establishing the optimization problem of JSSDL, we design an optimization method to solve it. Intuitively, (9) can be solved via an alternate iterative optimization method. However, since it involves a kernel norm of $D_2$, it cannot be solved directly. Therefore, an auxiliary matrix $P$ is introduced in place of $D_2$ in the low-rank constraint [26] as follows:

$$\min_{D_1,D_2,X_1,X_2,P}\left\{\begin{array}{l}\|Y-D_1X_1-D_2X_2\|_F^2+\lambda_1\|X_1\|_1\\ \quad+\lambda_2\|D_1\|_1+\lambda_3\|X_2\|_1+\lambda_4\|P\|_*\\ \quad+\lambda_5\|D_2-P\|_F^2\end{array}\right\}. \quad (10)$$

Thus, we separate the kernel norm and the Frobenius norm of $D_2$ and simplify the solving process. According to the alternate iterative optimization, the variables can be optimized one by one via fixing the others when optimizing one [22]. Accordingly, the proposed optimization algorithm includes four parts: specific dictionary updating, auxiliary matrix updating, shared dictionary updating, and sparse coding matrices updating. The detailed procedures are given as follows.

*a) Update specific dictionary $D_1$:* Fix unrelated variables, and let $Y_1 = Y - D_2X_2$. The optimization program can be obtained as follows:

$$D_1 = \arg\min_{D_1}\left\{\|Y_1 - D_1X_1\|_F^2 + \lambda_2\|D_1\|_1\right\}. \quad (11)$$

Let $D_1 = [d_1,\ldots,d_j,\ldots,d_k]$. Correspondingly, $X_1{=}[{x_T^1}^T,\ldots,{x_T^j}^T,\ldots,{x_T^k}^T]^T$, where $x_T^j$ denotes the row vector of the $j$th row in $X_1$. Thus, (11) can be expressed as follows:

$$\left\|Y_1-\sum_{i=1}^{k}d_ix_T^i\right\|_F^2+\sum_{i=1}^{k}\|d_i\|_1. \quad (12)$$

Solve for $D_1$ column by column. When solving for the $j$th atom, $d_j$ is listed separately, and (12) can be rewritten as follows:

$$\left\|Y_1-\sum_{i\neq j}d_ix_T^i-d_jx_T^j\right\|_F^2+\left\|d_j\right\|_1. \quad (13)$$

Assuming that $E_j = Y_1 - \sum_{i\neq j} d_ix_T^i$ is the residual matrix, the optimization problem can be described as follows:

$$\min_{d_j}\ \left\|E_j-d_jx_T^j\right\|_F^2+\left\|d_j\right\|_1 \quad (14)$$

where $d_j$ should be sparse, so most of the row vectors in $d_jx_T^j$ are zero vectors. Let $d_j = [d_{j1}, d_{j2},\ldots,d_{jm}]^T$. $d_{ji}$ denotes the $i$th element of the column vector $d_j$, $i \in [1,m]$. Accordingly, $E_j = [e_{j1}^T, e_{j2}^T,\ldots,e_{jm}^T]^T$. $e_{ji}$ denotes the $i$th row vector in $E_j$, $i\in[1,m]$.

Set the sparsity of $d_j$ as $a$, and solve each element of $d_j$. Let $L_1 = [\|e_{j1}\|_1,\ldots,\|e_{ji}\|_1,\ldots,\|e_{jm}\|_1]$. Here, we define a sort function $f = \mathrm{Sort}(x)$, where $x$ is an element of matrix $X$ and $f$ is the sorting number of $x$ in the order of $X$ from largest to smallest. When $\mathrm{Sort}(\|e_{ji}\|_1) \le a$, the corresponding $d_{ji}$ takes a nonzero value. The rest $d_{ji}$ takes as 0.

When $d_{ji}$ takes a nonzero value, we need to calculate it. The following equation is satisfied:

$$e_{ji} \approx d_{ji}x_T^j. \quad (15)$$

Try to minimize the residual vector $\vec{e}$ between $d_{ji}x_T^j$ and $e_{ji}$. As the value of $d_{ji}$ changes, the length of $d_{ji}x_T^j$ changes, causing the magnitude and direction of the residual vector $\vec{e}$ to change as well. According to Fig. 3, $\vec{e}$ is minimum when the direction of $\vec{e}$ is perpendicular to $x_T^j$. It is clear that $d_{ji}$ at this situation is the most appropriate.

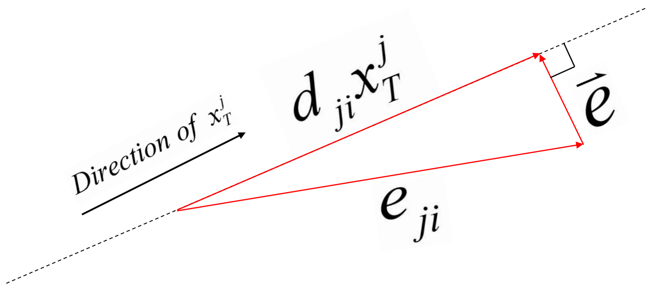



Fig. 3. Geometric interpretation of $d_{ji}x_T^j$ and $e_{ji}$.

To summarize, use the following equation to solve for each element $d_{ji}$ of $d_j$:

$$d_{ji} = \begin{cases} \frac{\langle e_{ji}\cdot x_T^j \rangle}{\left\| x_T^j \right\|_2^2}, & \text{if } \operatorname{Sort}(\left\| e_{ji} \right\|_1) \le a \\ 0, & \text{otherwise.} \end{cases} \quad (16)$$

*b) Update auxiliary matrix P:* Ignore irrelevant terms to $P$; the objective function can be rewritten as follows:

$$P = \arg\min_P \left\{ \lambda_4 \|P\|_* + \lambda_5 \|D_2 - P\|_F^2 \right\}. \quad (17)$$

First, the soft threshold function is defined as follows:

$$S_\tau(\sigma) = \operatorname{sign}(\sigma) \cdot \max(\sigma - \tau, 0) \quad (18)$$

where $\tau$ denotes the soft threshold and $\sigma$ denotes the input of the function. When the input of the soft threshold function is a matrix, the solution is using the soft threshold function for each element to obtain the corresponding matrix. Based on singular value decomposition, the update formula for $P$ is obtained as follows:

$$D_2 = SUV^T; \;\; P = S \cdot S_\tau(U) \cdot V^T. \quad (19)$$

In (19), the diagonal matrix $U$ and the two orthogonal matrices $S$ and $V$ are the results of the singular value decomposition of $D_2$, where on the main diagonal of $U$ are the singular values in descending order. When the threshold $\tau$ is chosen appropriately, $S_\tau(U)$ can approximate $U$ as much as possible to ensure the minimization of the regular term $\|D_2 - P\|_F^2$. With iterations, the smaller singular values in matrix $S_\tau(U)$ gradually shrink to 0, and $S_\tau(U)$ becomes a low-rank matrix, ensuring the minimization of the regular term $\|P\|_*$, i.e., $P$ gradually becomes a low-rank matrix.

*c) Update shared dictionary $D_2$:* Ignore irrelevant terms to $D_2$; the objective function regarding to $D_2$ can be rewritten as follows:

$$D_2 = \arg\min_{D_2} \left\{ \|Y - D_1X_1 - D_2X_2\|_F^2 + \lambda_5 \|D_2 - P\|_F^2 \right\}. \quad (20)$$

Since the Frobenius norm is convex, we can obtain the closed-form solution of $D_2$ directly by calculating the partial derivative

$$D_2 = \left((Y - D_1X_1)X_2^T + \lambda_5 P\right) \cdot \left(X_2X_2^T + \lambda_5 E\right)^{-1}. \quad (21)$$

*d) Update sparse coding $X_1$ and $X_2$:* The objective function turns to the following form when keeping the terms relevant only to $X_1$ and $X_2$:

$$\{X_1, X_2\} = \arg\min_{X_1,X_2} \left\{ \begin{array}{l} \|Y - D_1X_1 - D_2X_2\|_F^2 \\ + \lambda_1\|X_1\|_1 + \lambda_3\|X_2\|_1 \end{array} \right\}. \quad (22)$$

There are two sparse coding matrices required to be solved, which cannot be solved directly by the OMP method, so $X_1$ and $X_2$ are solved by three steps as follows:

$$X_2 = \arg\min_{X_2} \left\{ \|Y - D_2X_2\|_F^2 + \lambda_3\|X_2\|_1 \right\} \quad (23)$$
$$X_1 = \arg\min_{X_1} \left\{ \|Y - D_2X_2 - D_1X_1\|_F^2 + \lambda_1\|X_1\|_1 \right\} \quad (24)$$
$$X_2 = \arg\min_{X_2} \left\{ \|Y - D_1X_1 - D_2X_2\|_F^2 + \lambda_3\|X_2\|_1 \right\}. \quad (25)$$

According to the assumptions, $Y$ consists of characteristics and commonalities. In (23), shared dictionary $D_2$ is first used separately to reconstruct $Y$ and calculate $X_2$ because the commonalities are the main component of $Y$. Then, in (24), the specific dictionary $D_1$ is used to reconstruct characteristics and calculate $X_1$, where characteristics can be denoted as $Y - D_2X_2$. Likewise, in (25), commonalities can be denoted as $Y - D_1X_1$, and use $D_2$ to reconstruct it. Through the above three steps, the specific coding matrix and the shared coding matrix are solved separately to better extract the commonalities and characteristic features of the data. Apparently, (23)–(25) can be solved by the OMP method separately. First, precompute $X_2$ by (21), then calculate $X_1$ by (22), and calculate $X_2$ by (23).

In summary, according to the alternate iterative optimization method, the specific dictionary, i.e., the shared dictionary, can be obtained. In the optimization algorithm, we first solve for $D_1$, $P$, $D_2$, and, finally, sparse coding matrices. According to the theory of the alternate iterative optimization method [27], different iteration sequences do not affect the final results, all of which can cause the results to converge to the optimal solution. The above four parts are iterated sequentially until convergence or stopping criteria are reached. The optimization problem of (9) includes four optimization variables: $D_1$, $D_2$, $X_1$, and $X_2$. Although (9) is not joint convex to ($D_1$, $D_2$, $X_1$, and $X_2$), it is convex with respect to each of them when the others are fixed. Therefore, the convergence property of the proposed optimization method can be guaranteed, the convergence of which has been studied in [25] and [28]. The detailed steps about the optimization algorithm of JSSDL are summarized in Algorithm 1.

*2) Online Process Monitoring:* After obtaining the dictionaries from the training data, the reconstruction errors of the training data can be calculated as follows:

$$R_j = \left\| y_j - D_1x_{1j} - D_2x_{2j} \right\|_2^2 \quad (26)$$

where $R_j$ denotes the reconstruction error of the $j$th training data, and $x_{1j}$ and $x_{2j}$ denote the sparse coding of the $j$th characteristic and commonality. After calculating the reconstruction errors of training data, the threshold $R_{\text{tr}}$ can be obtained by the kernel density estimation (KDE) method [29],

**Algorithm 1** Jointly Specific and Shared Dictionary Learning

**Input**: Date $Y$, sizes of $D_1$ and $D_2$, the sparsity of $D_1$, the sparsity of $X_1$ and $X_2$.
**Initialization:** $D_1$, $D_2$,$X_1$ and $X_2$ randomly
**While** $k < iteration$ **do**
**Step 1: Update specific dictionary.** Fix $D_2(k)$, $X_1(k)$, $X_2(k)$ and $P(k)$
to obtain $D_1(k+1)$ according to Eq. (16);
**Step 2: Update low-rank auxiliary matrix.** Fix $D_1(k+1)$, $D_2(k)$, $X_1(k)$
and $X_2(k)$ to obtain $P(k+1)$ according to Eq. (19);
**Step 3: Update shared dictionary.** Fix $D_1(k+1)$, $X_1(k)$, $X_2(k)$ and
$P(k+1)$ to obtain $D_2(k+1)$ according to Eq. (21);
**Step 4: Update sparse coding matrices.** Fix $D_1(k+1)$, $D_2(k+1)$
and $P(k+1)$ to obtain $X_1(k+1)$ and $X_2(k+1)$ according to Eqs. (23)-(25);
**Step 5: Record the number of iterations.** $k = k+1$;
**end while**
**Output**: Specific dictionary $D_1$ and shared dictionary $D_2$

which is defined as follows:

$$f(R) = \frac{1}{Mh}\sum_{i=1}^{M} K\left(\frac{|R - R_j|}{h}\right) \tag{27}$$

where $M$ denotes the number of training samples and $h$ denotes the bandwidth. $f(R)$ is the density function, and $K(x)$ denotes the Gaussian kernel function. The probability density function of the reconstruction error is obtained according to (27). Then, choose the appropriate confidence level $\alpha$ to calculate the Confidence interval of the reconstruction error. The upper limit of the confidence interval is the threshold $R_{\text{tr}}$, which serves as a control limit for reconstruction error. When the reconstruction error of test data is greater than $R_{\text{tr}}$, the data are faulty.

When online data arrive, we use the learned specific dictionary and shared dictionary to reconstruct the newly arrived data. Mathematically,

$$\begin{aligned}\{x_{\text{new}1}, x_{\text{new}2}\} = \arg\min_{x_1,x_2} \|y_{\text{new}} - D_1x_1 - D_2x_2\|_2^2 \\ \text{s.t. } \|x_1\|_0 \le T_1,\ \|x_2\|_0 \le T_2.\end{aligned} \tag{28}$$

After obtaining the sparse coding, the joint reconstruction error (JRE) is calculated by the following equation:

$$\text{JRE}_{\text{new}} = \|y_{\text{new}} - D_1x_{\text{new}1} - D_2x_{\text{new}2}\|_2^2. \tag{29}$$

Intuitively, if the data sample is normal, it can be well represented by the specific dictionary and shared dictionary; thus, its JRE will be less than the threshold. If the data sample is faulty, the JRE will be large than $R_{\text{tr}}$ because the dictionaries cannot represent faulty data. Therefore, we determine the state of the new data by comparing $\text{JRE}_{\text{new}}$ with the threshold $R_{\text{tr}}$. If $\text{JRE}_{\text{new}} > R_{\text{tr}}$, the new test sample is considered as faulty data. Otherwise, the data are considered normal.

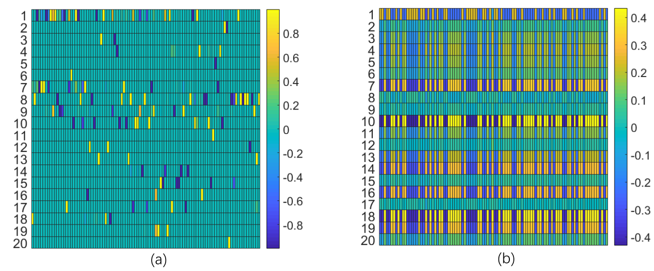



Fig. 4. (a) Heat map of $D_1$. (b) Heat map of $D_2$.

## III. Illustrative Examples

In this section, three examples, including numerical simulation, TE process, and roasting process, are introduced to verify the effectiveness of the proposed method. In order to assess the process monitoring performance quantitatively, FDR and FAR are defined as follows:

$$\text{FDR} = \frac{\text{tp}}{\text{total count of faulty data}} \times 100\% \tag{30}$$

$$\text{FAR} = \frac{\text{fp}}{\text{total count of normal data}} \times 100\% \tag{31}$$

where tp and fp denote the number of correctly judged fault samples and misjudged normal samples. In addition, to verify the superiority of the proposed method, some state-of-the-art methods, including the PCA method [30], the robust PCA method [31], and the traditional dictionary learning method [25], are introduced for comparison.

### *A. Numerical Simulation Example*

In order to verify the performance of the proposed method, we designed a numerical simulation experiment. Numerical data are generated as follows:

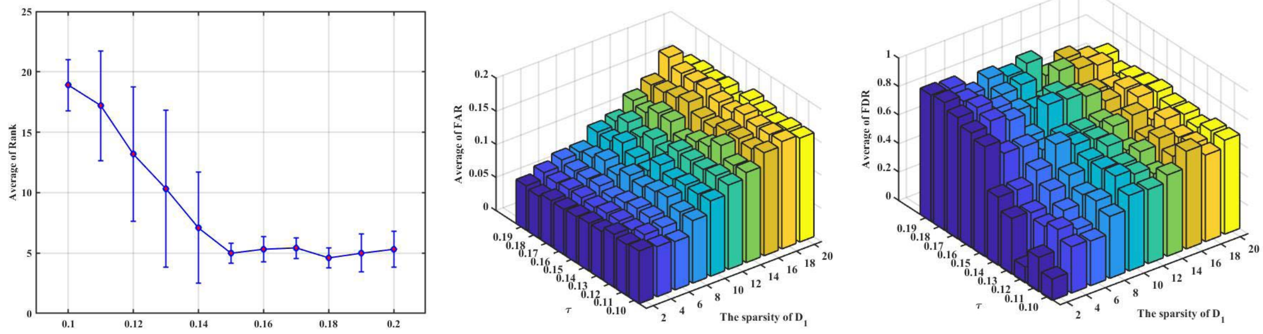
$$y = k_1y_1 + k_2y_2 + e \tag{32}$$

where $k_i \sim N(1,1)$, $i = 1,2$. $e = [e_1,\ldots,e_i,\ldots,e_{20}]^T$ denotes the Gaussian noise, which satisfies $e_i \sim N(0,0.1)$. According to the assumptions, we generated $y_1$ and $y_2$ as follows to ensure the sparse feature of characteristics and strongly correlated feature of commonalities:

$$y_1 = k_1i_a + k_2i_b; \quad y_2 = As \tag{33}$$

where $k_i \sim N(1,1)$, $i = 1,2$. $i_a$ and $i_b$ are the column vectors in a unit matrix. $A \in R^{20\times 2}$ denotes the observation matrix of commonalities. $\text{s} = [s_1, s_2]^T$ is the state matrix and satisfies $s_i \sim N(1,0.5)$, $i = 1,2$. Hereafter, the generated data are used to verify the validity of the proposed method.

First, to verify that the proposed optimization algorithm can train a low-rank dictionary and a sparse dictionary, we generated numerical data and trained $D_1, D_2 \in R^{20\times 100}$. $D_1$ and $D_2$ are visualized in Fig. 4. In Fig. 4(a), there are only a few nonzero values in each column, which indicates that the optimization algorithm can train a sparse dictionary. In Fig. 4(b), most of the columns are the same, indicating that the $D_2$ is a low-rank matrix. Thus, we can conclude that the




(a) (b) (c)

Fig. 5. (a) Effect of the soft threshold on the rank of $D_2$. (b) Effect of $\tau$ and the sparsity of $D_1$ on FAR. (c) Effect of $\tau$ and the sparsity of $D_1$ on FDR.

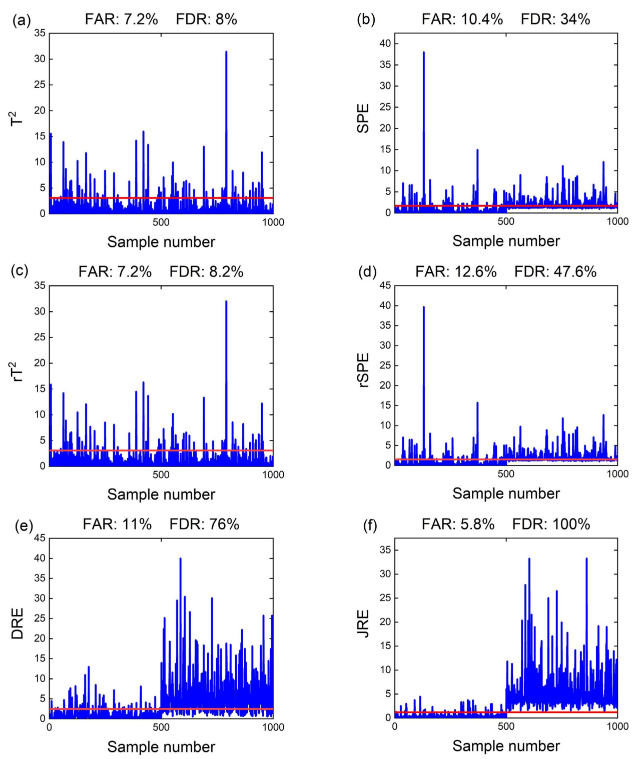



Fig. 6. Experiment results of the numerical example. (a) $T^2$ statistic of the PCA method. (b) SPE statistic of the PCA method. (c) $T^2$ statistic of the robust PCA method. (d) SPE statistic of the robust PCA method. (e) DRE statistic of dictionary learning. (f) JRE statistic of the proposed method.

optimization algorithm can ensure the sparsity and low-rank features of $D_1$ and $D_2$.

After verifying the effectiveness of the optimization algorithm, the process monitoring procedure can be performed in three steps.

1) *Data Generation:* Generate training data and test data by (32). For the training data, we generate 1000 normal samples. For the test data, it consisted of two parts: 500 normal data and 500 faulty data. Faulty data are the normal data with a bias fault added in the seventh dimension $y_7$.
2) *Model Training:* Use the training data to learn the specific dictionary and shared dictionary based on the designed optimization algorithm. Then, the reconstruction error of training data can be calculated. Finally, the threshold of reconstruction error can be calculated based on the KDE method.
3) *Online Decision-Making:* After obtaining the two dictionaries, the sparse coding of the test data is obtained, and the reconstruction error of the test data is calculated. By comparing the reconstruction error JRE with the threshold $R_{\text{tr}}$, we can obtain the state of test data.

Considering that the sparsity of $D_1$ and the rank of $D_2$ are two important parameters of the proposed JSSDL method, we conduct a parameter-sensitive analysis experiment to examine their effects on FAR and FDR. We vary the sparsity of $D_1$ directly from 2 to 20, with 2 per change. For the rank of $D_2$, which cannot be adjusted directly in the optimization algorithm, we adjust it by changing the soft threshold $\tau$ and, thus, further change the rank of $D_2$. We change $\tau$ from 0.1 to 0.19 without loss of generality. In this experiment, the number of atoms of both $D_1$ and $D_2$ is 100, the sparsity of $X_1$ is 2, the sparsity of $X_2$ is 3, and the confidence level of the KDE is 0.92. A bias fault of +8 is added to $y_7$ of the last 500 test data. To eliminate the fluctuation induced by the random initialization, we did several experiments and calculated the mean of FAR and FDR to judge the effectiveness of process monitoring. Fig. 5 shows the results. In Fig. 5(a), the rank of $D_2$ decreases with the increase in $\tau$, which verifies the effectiveness of using soft threshold iterations in the optimization algorithm and illustrates the reasonableness of adjusting $\tau$ to change the rank of $D_2$. In Fig. 5(b) and (c), we obtained the variation pattern of FAR and FDR. As the sparsity of $D_1$ increases, the FAR keeps increasing, indicating that process monitoring performance becomes worse. The change of the rank of $D_2$ has no significant effect on the FAR. In Fig. 5(c), when $\tau$ is relatively large, and the sparsity of $D_1$ is small, the FDR is relatively large. This combination of parameters enables JSSDL to better learn the commonality and characteristic features of the data, and therefore, the FDR is significantly improved. This result indicates the combination of the low-rank dictionary, and sparse dictionary is reasonable and can improve the process monitoring effect. In addition, the best combination of parameters on the numerical simulation data that $\tau$ is 0.18, and the sparsity of $D_1$ is 2 is obtained and used in the next experiment.

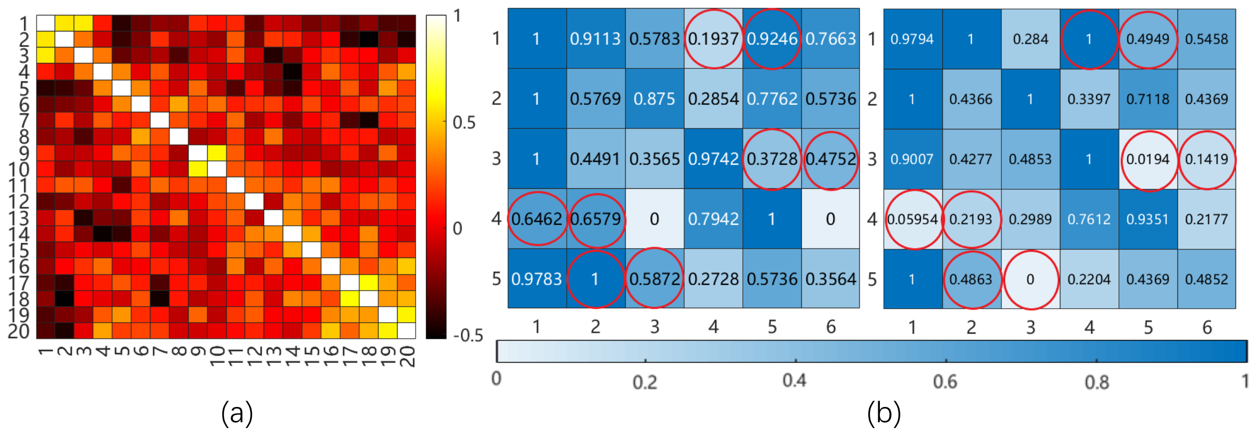



Fig. 7. (a) Correlation coefficient matrix of 20 TE process samples. (b) Visualization of the first 30 dimensions of two normalized samples.

Next, we compare the process monitoring performance of the proposed method with three state-of-the-art methods based on the numerical simulation data. The parameters of each method are set as follows. In PCA and robust PCA, the CPV is 0.9. The regularization parameters of the robust PCA method are set according to its original settings: $\lambda = (1/\sqrt{\max(m, M)})$ and $\mu = 10\lambda$. Therefore, $\lambda = 0.0316$ and $\mu = 0.316$. For dictionary learning, the number of atoms is 100, and the sparsity of the sparse coding is 3. For the proposed method, the number of atoms of both $D_1$ and $D_2$ is 100, the sparsity of $X_1$ is 2, and the sparsity of $X_2$ is 3. According to the previous experiment, the sparsity of $D_1$ is 2, and $\tau$ is 0.18. The confidence level of KDE of all methods is 0.92 to ensure a fair comparison. A bias fault of +10 is added to $y_7$ of the last 500 test data. Fig. 6 shows the quantitative results of all methods. Since the data of numerical simulation have obvious characteristics and commonalities features, it is difficult for the PCA method and the robust PCA method to monitor such data with complex features. Therefore, the $T^2$ and SPE statistics of PCA, as well as robust PCA, cannot obtain a satisfactory FDR. The dictionary learning is effective than the PCA and robust PCA method, However, due to its relatively poor representation of normal data, a larger FAR was created, and the FDR was affected to some extent. Among these four methods, the proposed method achieves the optimum performance.

## B. TE Process Example

The TE benchmark process is a simulation of real industrial processes created by Eastman Chemical Company and is the most widely used model for evaluating process monitoring methods [32]–[35]. Due to the complex feature of the TE process, the data obtained in the TE process are consistent with the assumptions of this article, and we chose it to validate the process monitoring performance of the proposed method.

In the TE process, there are 41 measured variables (22 continuous process variables and 19 composition variables) and 12 manipulated variables. A total of 21 faults were introduced during the process. Similar to [36] and [37], 22 continuous process variables and 11 manipulated variables were used in this experiment. Detailed data description is referenced to [38]. A total of 22 datasets are used in this experiment: the dataset of the normal mode as the training set and the datasets of 21 fault modes as the test sets. All faults in the test datasets were introduced after the 160th sample.

First, to explain the characteristics and commonalities feature of the TE process data, we performed a correlation analysis. The correlation coefficient matrix of 20 training samples is shown in Fig. 7(a). It can be seen that the correlations between most of the samples are not strong. To further investigate the underlying reason, we visualize the first 30 dimensions of two normalized samples in Fig. 7(b). When the difference in the same dimension at different moments is greater than 0.3, we consider it to be characteristic in this dimension and circle it. Apparently, there are eight dimensions with characteristics in total, which causes a weak correlation between the data. Since the data satisfy the assumptions that there are characteristics and commonalities in data, the proposed method can obtain a good performance.

Next, we compare the monitoring effectiveness of different methods. The parameters of different methods are set fairly as follows. For the proposed method, the sparsities of both $X_1$ and $X_2$ are 5, and the sparsity of $D_1$ is 5. The number of atoms for both $D_1$ and $D_2$ is 100. $\tau$ is 0.08. For the dictionary learning method, the number of atoms is 100, and the sparsity is 3. For the PCA and robust PCA methods, the CPV is 0.9. For the robust PCA method, $\lambda = 0.0447$, and $\mu = 0.447$. The confidence level of all methods is 0.99. The FDR of 21 faults is shown in Table I. Apparently, the FDR of the proposed method is higher than other methods in the majority of faults. When faults 5, 16, 19, and 20 occur, it is difficult for other methods to detect faulty data, but the proposed method still has a relatively high FDR. The average value of FDR further verified the superiority of the proposed method. Due to space limitations, the vivid process monitoring results of the proposed method of faults 10, 13, 19, and 20 are shown in Fig. 8.

## C. Roasting Process Experiment

To further verify the applicability of the proposed method, an industrial roasting process of a zinc smelting factory in Hunan province, China, was introduced. Roasting is the first step in the zinc smelting process. The stable and safe operation

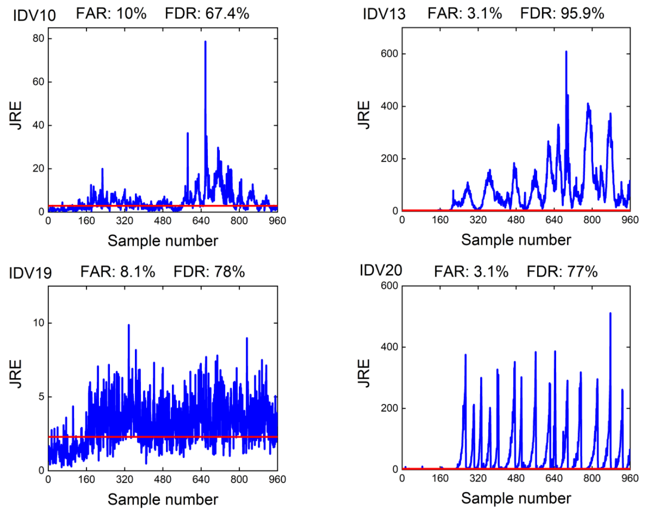



Fig. 8. Monitoring results of the proposed method for faults 10, 13, 19, and 20.

TABLE I
FDR OF TE PROCESS

| Fault | PCA method $T^2$ | PCA method $SPE$ | robust PCA method $rT^2$ | robust PCA method $rSPE$ | DL method | JSSDL method |
|---|---|---|---|---|---|---|
| 1 | 99.3% | **100%** | 99.3% | **100%** | 99.1% | 99.5% |
| 2 | 98.2% | **99.4%** | 98.4% | 99.3% | 96.8% | 99% |
| 3 | 5.8% | 4.9% | 10.8% | 6.1% | 6.6% | **18.8%** |
| 4 | 68.1% | 100% | 74.1% | **100%** | 1.8% | 63.9% |
| 5 | 27.8% | 29.4% | 30.9% | 33.8% | 24.5% | **64.1%** |
| 6 | 99.5% | **100%** | 99.9% | **100%** | 99.9% | 99.8% |
| 7 | **100%** | **100%** | **100%** | **100%** | 84.3% | 97.6% |
| 8 | 97.2% | 95.8% | 97.5% | 96.1% | 91.6% | **98.1%** |
| 9 | 5.6% | 4.8% | 8.6% | 6.3% | 6.6% | **17%** |
| 10 | 44.5% | 59.4% | 48.8% | 61.4% | 29.4% | **67.4%** |
| 11 | 60.7% | 66.5% | 64% | **73.3%** | 12% | 47.4% |
| 12 | 98.5% | 94.9% | 98.7% | 96.6% | 83.5% | **99%** |
| 13 | 94.4% | 95.6% | 94.4% | 95.6% | 93.3% | **95.9%** |
| 14 | **100%** | 99.3% | **100%** | 99.9% | 9.8% | 99.1% |
| 15 | 7.8% | 10.4% | 12.4% | 11.8% | 5.1% | **24%** |
| 16 | 29.8% | 53.5% | 36.2% | 57.3% | 27.9% | **74.9%** |
| 17 | 84.8% | 96.5% | 86.4% | **96.6%** | 19.5% | 76.1% |
| 18 | 89.6% | 90.6% | 90.4% | 90.8% | 87% | **92.6%** |
| 19 | 15.9% | 35.3% | 23.8% | 42.5% | 3.9% | **78%** |
| 20 | 43% | 64.4% | 48.8% | 67.1% | 37.6% | **77%** |
| 21 | 43.5% | 57% | 47.3% | 57.5% | 37.5% | **57.9%** |
| Average | 62.6% | 69.4% | 65.3% | 71.0% | 45.6% | **73.7%** |

of the roasting process is highly important to reduce industrial pollution and ensure the quality of the output zinc. The main function of the roasting process is to oxidize the raw ore at high temperature in the roaster so that the insoluble zinc sulfide is converted into zinc oxide soluble in weak acids, and the output material zinc roast is transported to the subsequent leaching process. The structure of the roasting process is shown in Fig. 9. It makes sense to perform process monitoring for the roasting process to ensure a stable and safe operation.

In the roasting process, a total of 25 dimensions of variables were measured, including the temperature measurements in the roaster, the feed volume, blowing pressure and rate, outlet flue gas pressure and temperature, and the

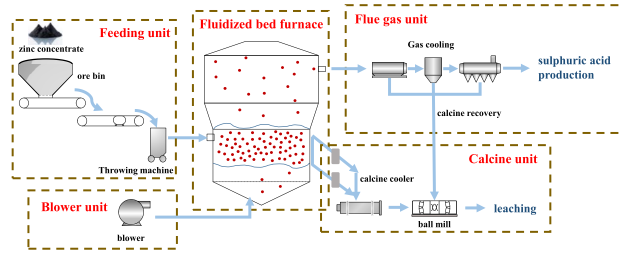



Fig. 9. Structure of the roasting process.

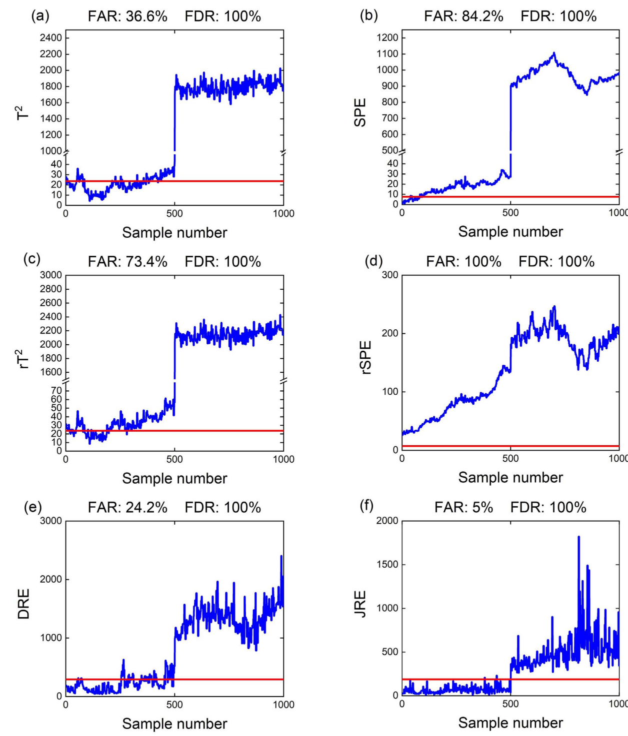



Fig. 10. Process monitoring results of the roasting process. (a) $T^2$ statistic of the PCA method. (b) SPE statistic of the PCA method. (c) $T^2$ statistic of the robust PCA method. (d) SPE statistic of the robust PCA method. (e) DRE statistic of dictionary learning. (f) JRE statistic of the proposed method.

$SO_2$ concentration. The detailed physical significance of each dimension is given in Table II; 1000 normal samples were used as training data, and 500 normal samples and 500 samples for the faulty underoxidation condition were used as test data.

We compare the monitoring effectiveness of different methods. For a fair comparison, the parameters of different methods are set as follows. For the proposed method, the sparsities of $X_1$ and $X_2$ are 4 and 2, and the sparsity of $D_1$ is 2. The number of atoms for both $D_1$ and $D_2$ is 100. $\tau$ is 0.01. For the dictionary learning method, the number of atoms is 100, and the sparsity is 3. For the PCA and robust PCA method, the CPV is 0.9. For the robust PCA method, $\lambda = 0.0316$ and $\mu = 0.316$. The confidence level of all methods is 0.99. Fig. 10 shows the experiment results. It can be seen that the faulty working condition can be well detected by various methods. However, since the proposed method can extract the characteristics feature perfectly, its FAR is better than

TABLE II
MONITORING VARIABLES OF THE ROASTING PROCESS

| No. | Monitoring variable | No. | Monitoring variable | No. | Monitoring variable | No. | Monitoring variable |
|---|---|---|---|---|---|---|---|
| 1 | Actual value of feed volume | 8 | Temperature A of the upper part of the boiling layer | 14 | Temperature B in the middle of the boiling layer | 20 | Temperature C of the lower part of the boiling layer |
| 2 | Blower current | 9 | Temperature B of the upper part of the boiling layer | 15 | Temperature C in the middle of the boiling layer | 21 | Temperature D of the lower part of the boiling layer |
| 3 | Roaster inlet air main duct flow | 10 | Temperature C of the upper part of the boiling layer | 16 | Temperature D in the middle of the boiling layer | 22 | Temperature E of the lower part of the boiling layer |
| 4 | Roaster inlet air main duct flow pressure | 11 | Temperature D of the upper part of the boiling layer | 17 | Temperature E in the middle of the boiling layer | 23 | Temperature of roaster outlet flue gas |
| 5 | Roaster inlet air duct temperature | 12 | Temperature E of the upper part of the boiling layer | 18 | Temperature A of the lower part of the boiling layer | 24 | Pressure of roaster outlet flue gas |
| 6 | Roaster chamber temperature A | 13 | Temperature A in the middle of the boiling layer | 19 | Temperature B of the lower part of the boiling layer | 25 | SO2 concentration |
| 7 | Roaster chamber temperature B | | | | | | |

other methods, which further indicates the effectiveness of the proposed method.

## IV. Conclusion and Discussion

Considering that raw data of industrial processes are always featured by characteristics and commonalities, which raises difficulties in process monitoring, this article proposes a JSSDL method to obtain an accurate process monitoring performance. First, we build the specific dictionary and shared dictionary to extract the characteristics and commonalities feature of raw data. Then, the control limit for process monitoring is obtained from the training data by using the KDE method. When new data arrive, the data are reconstructed by the two dictionaries, and the JRE is compared with the control limit for process monitoring. To illustrate the effectiveness of the method, extensive experiments, including numerical simulation experiments, TE process experiments, and an industrial roasting process experiment, are conducted. Compared with some state-of-the-art methods, the proposed method can learn the feature of the data better, have better data representation capability, and, thus, further achieve satisfactory process monitoring results. The proposed method is applicable to complex industrial process monitoring, but we have only studied process monitoring in a single mode at present. It is an interesting future research direction to extend the method to multimode process monitoring. In addition, since the data of different modes may have different commonalities and characteristic features, if these discriminative features can be extracted, it will be very helpful for the mode identification of process data, which also will be investigated in our future work.

## References

[1] G. Mois, T. Sanislav, and S. C. Folea, "A cyber-physical system for environmental monitoring," *IEEE Trans. Instrum. Meas.*, vol. 65, no. 6, pp. 1463–1471, Jun. 2016.

[2] Z. Asad, M. A. R. Chaudhry, and D. Kundur, "On the use of Matroid theory for distributed cyber–physical-constrained generator scheduling in smart grid," *IEEE Trans. Ind. Electron.*, vol. 62, no. 1, pp. 299–309, Jan. 2015.

[3] Y. Li, E. Zio, N. Lu, X. Wang, and B. Jiang, "Joint distribution-based test selection for fault detection and isolation under multiple faults condition," *IEEE Trans. Instrum. Meas.*, vol. 70, pp. 1–13, 2021.

[4] Z. Guo, Y. Wan, and H. Ye, "An unsupervised fault-detection method for railway turnouts," *IEEE Trans. Instrum. Meas.*, vol. 69, no. 11, pp. 8881–8901, Nov. 2020.

[5] K. Huang, H. Wen, C. Zhou, C. Yang, and W. Gui, "Transfer dictionary learning method for cross-domain multimode process monitoring and fault isolation," *IEEE Trans. Instrum. Meas.*, vol. 69, no. 11, pp. 8713–8724, Nov. 2020.

[6] X. Ma, Y. Si, Z. Yuan, Y. Qin, and Y. Wang, "Multistep dynamic slow feature analysis for industrial process monitoring," *IEEE Trans. Instrum. Meas.*, vol. 69, no. 12, pp. 9535–9548, Dec. 2020.

[7] K. Huang, S. Wu, F. Li, C. Yang, and W. Gui, "Fault diagnosis of hydraulic systems based on deep learning model with multirate data samples," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Jun. 10, 2021, doi: 10.1109/TNNLS.2021.3083401.

[8] L. Feng and C. Zhao, "Fault description based attribute transfer for zero-sample industrial fault diagnosis," *IEEE Trans. Ind. Informat.*, vol. 17, no. 3, pp. 1852–1862, Mar. 2021.

[9] S. Zhang and C. Zhao, "Slow-feature-analysis-based batch process monitoring with comprehensive interpretation of operation condition deviation and dynamic anomaly," *IEEE Trans. Ind. Electron.*, vol. 66, no. 5, pp. 3773–3783, May 2019.

[10] Q. Jiang, X. Yan, and B. Huang, "Performance-driven distributed PCA process monitoring based on fault-relevant variable selection and Bayesian inference," *IEEE Trans. Ind. Electron.*, vol. 63, no. 1, pp. 377–386, Jan. 2016.

[11] Q. Jiang *et al.*, "Neural network aided approximation and parameter inference of non-Markovian models of gene expression," *Nature Commun.*, vol. 12, no. 1, pp. 1–12, Dec. 2021.

[12] Y. Si, Y. Wang, and D. Zhou, "Key-performance-indicator-related process monitoring based on improved kernel partial least squares," *IEEE Trans. Ind. Electron.*, vol. 68, no. 3, pp. 2626–2636, Mar. 2021.

[13] Z. Chen *et al.*, "A just-in-time-learning-aided canonical correlation analysis method for multimode process monitoring and fault detection," *IEEE Trans. Ind. Electron.*, vol. 68, no. 6, pp. 5259–5270, Jun. 2021.

[14] Y. Dong and S. J. Qin, "A novel dynamic PCA algorithm for dynamic data modeling and process monitoring," *J. Process Control*, vol. 67, pp. 1–11, Jul. 2018.

[15] Q. Jiang, S. Yan, H. Cheng, and X. Yan, "Local-global modeling and distributed computing framework for nonlinear plant-wide process monitoring with industrial big data," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 8, pp. 3355–3365, Aug. 2021.

[16] M. Elad, M. A. T. Figueiredo, and Y. Ma, "On the role of sparse and redundant representations in image processing," *Proc. IEEE*, vol. 98, no. 6, pp. 972–982, Jun. 2010.

[17] Z. Jiang, Z. Lin, and L. S. Davis, "Label consistent K-SVD: Learning a discriminative dictionary for recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 35, no. 11, pp. 2651–2664, Nov. 2013.

[18] J. Mairal, F. Bach, J. Ponce, G. Sapiro, and A. Zisserman, "Supervised dictionary learning," 2008, *arXiv:0809.3083*.

[19] Q. Zhang and B. Li, "Discriminative K-SVD for dictionary learning in face recognition," in *Proc. 23rd IEEE Comput. Vis. Pattern Recognit.*, San Francisco, CA, USA, Jun. 2010, pp. 13–18.

[20] L. Ren and W. Lv, "Fault detection via sparse representation for semiconductor manufacturing processes," *IEEE Trans. Semicond. Manuf.*, vol. 27, no. 2, pp. 252–259, May 2014.

[21] K. Huang, Y. Wu, C. Yang, G. Peng, and W. Shen, "Structure dictionary learning-based multimode process monitoring and its application to aluminum electrolysis process," *IEEE Trans. Autom. Sci. Eng.*, vol. 17, no. 4, pp. 1989–2003, Oct. 2020.

[22] K. Huang, Y. Wu, C. Wang, Y. Xie, C. Yang, and W. Gui, "A projective and discriminative dictionary learning for high-dimensional process monitoring with industrial applications," *IEEE Trans. Ind. Informat.*, vol. 17, no. 1, pp. 558–568, Jan. 2021.

[23] C. Yang *et al.*, "Multimode process monitoring based on robust dictionary learning with application to aluminium electrolysis process," *Neurocomputing*, vol. 332, pp. 305–319, Mar. 2019.

[24] S. K. Sahoo and A. Makur, "Signal recovery from random measurements via extended orthogonal matching pursuit," *IEEE Trans. Signal Process.*, vol. 63, no. 10, pp. 2572–2581, May 2015.

[25] M. Aharon, M. Elad, and A. Bruckstein, "K-SVD: An algorithm for designing overcomplete dictionaries for sparse representation," *IEEE Trans. Signal Process.*, vol. 54, no. 11, pp. 4311–4322, Nov. 2006.

[26] Y.-Q. Zhao and J. Yang, "Hyperspectral image denoising via sparse representation and low-rank constraint," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 1, pp. 296–308, Jan. 2015.

[27] C. Hu, G. Wang, K. C. Ho, and J. Liang, "Robust ellipse fitting with Laplacian kernel based maximum correntropy criterion," *IEEE Trans. Image Process.*, vol. 30, pp. 3127–3141, 2021.

[28] J. Mairal, F. Bach, J. Ponce, and G. Sapiro, "Online dictionary learning for sparse coding," in *Proc. Int. Conf. Mach. Learn.*, 2009, pp. 689–696.

[29] S. J. Sheather and M. C. Jones, "A reliable data-based bandwidth selection method for kernel density estimation," *J. Roy. Statist. Soc. B, Methodol.*, vol. 53, no. 3, pp. 683–690, 1991.

[30] M. Kano, S. Hasebe, I. Hashimoto, and H. Ohno, "A new multivariate statistical process monitoring method using principal component analysis," *Comput. Chem. Eng.*, vol. 25, nos. 7–8, pp. 1103–1113, Aug. 2001.

[31] E. J. Candès and X. Li, "Robust principal component analysis?" *J. ACM*, vol. 58, no. 3, pp. 1–37, 2011.

[32] A. Raich and A. Cinar, "Multivariate statistical methods for monitoring continuous processes: Assessment of discrimination power of disturbance models and diagnosis of multiple disturbances," *Chemometrics Intell. Lab. Syst.*, vol. 30, no. 1, pp. 37–48, 1995.

[33] A. Singhal and D. E. Seborg, "Evaluation of a pattern matching method for the Tennessee eastman challenge process," *J. Process Control*, vol. 16, no. 6, pp. 601–613, Jul. 2006.

[34] L. H. Chiang, E. L. Russell, and R. D. Braatz, *Fault Detection and Diagnosis in Industrial Systems*. London, U.K.: Springer, 2001.

[35] Z. Ge and Z. Song, "Process monitoring based on independent component analysis–principal component analysis (ICA–PCA) and similarity factors," *Ind. Eng. Chem. Res.*, vol. 46, no. 7, pp. 2054–2063, Mar. 2007.

[36] G. Chen and Z. Ge, "Hierarchical Bayesian network modeling framework for large-scale process monitoring and decision making," *IEEE Trans. Control Syst. Technol.*, vol. 28, no. 2, pp. 671–679, Mar. 2020.

[37] S. Yin, S. X. Ding, A. Haghani, H. Hao, and P. Zhang, "A comparison study of basic data-driven fault diagnosis and process monitoring methods on the benchmark Tennessee Eastman process," *J. Process Control*, vol. 22, no. 9, pp. 1567–1581, 2012.

[38] A. Bathelt, N. L. Ricker, and M. Jelali, "Revision of the Tennessee Eastman process model," *IFAC-PapersOnLine*, vol. 48, no. 8, pp. 309–314, 2015.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